Rational Functions: Final Retake Practice: Parent Key

Worked solutions and proposed practice rubric. Keep separate from the student copy.

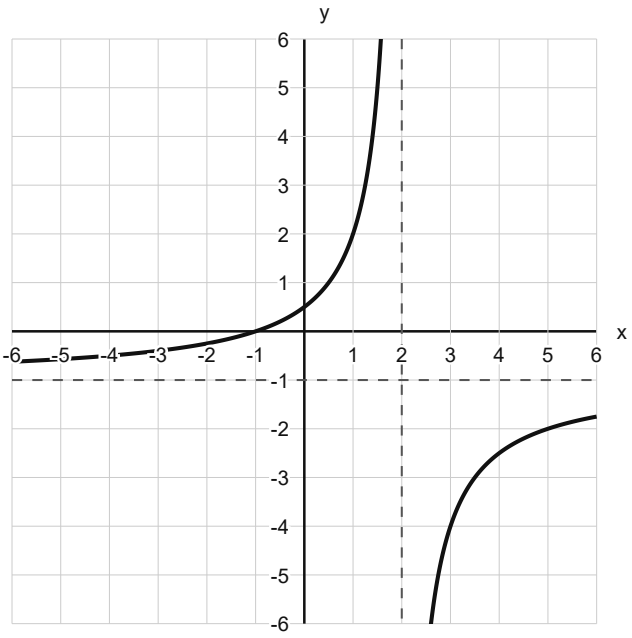
13 questions | 58 points | 75 minutes (user-selected allowance)

Award follow-through credit; do not repeatedly deduct an originating error. Accept equivalent valid functions and their own correct hole heights. This is not an official school key.

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1. 8 points

1. Rewrite $f(x) = -\dfrac{x+1}{x-2} = -1 - \dfrac{3}{x-2}$. Thus the vertical asymptote is $x = 2$ and the horizontal asymptote is $y = -1$.
2. For example, $f(1) = 2$, $f(3) = -4$, and $f(0) = 1/2$. Left of 2 the curve lies above -1; right of 2 it lies below -1.
3. As $x \to -\infty$, $f(x) \to -1$; as $x \to +\infty$, $f(x) \to -1$. The side of approach is useful but is not separately required for the end-limit points.



Practice rubric (8 points)

- 4: Graph: correct asymptote equations 1 (0.5 each), two correctly labeled points on different branches 1 (0.5 each), consistent left/right branches 2 (1 each). Do not penalize small freehand waviness as a new conceptual error.
- 2: Correct far-left input direction and output limit -1; 1 for the correct height if the input direction is missing or mislabeled.
- 2: Correct far-right input direction and output limit -1; 1 for the correct height if the input direction is missing or mislabeled.

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2. 4 points

1. Factor the denominators as $(x+3)(x-2)$ and $(x+3)(x+2)$. Their common denominator is $(x+3)(x-2)(x+2)$.
2. The numerator is $2(x+2)-3(x-2)=10-x$. No factor cancels. Retain the original exclusions $-3, -2, 2$.

Practice rubric (4 points)

- 1: Correct factorizations and common denominator; 0.5 for correctly factoring both but not completing the LCD.
- 2: Correct signed subtraction: 1 for setup 2(x+2)-3(x-2), 1 for numerator 10-x and complete fraction.
- 1: All original exclusions: one-third point each for -3, -2 and 2, with no spurious exclusion.

3. 4 points

1. Factor as $\dfrac{(x-3)(x+3)}{(x+2)(x+3)} \cdot \dfrac{(x+2)^2}{(x-3)^2}$.
2. Cancel only matching factors to leave $(x+2)/(x-3)$. The original denominator excludes $-3, -2, 3$. In particular -2 remains excluded even though the reduced numerator vanishes there.

Practice rubric (4 points)

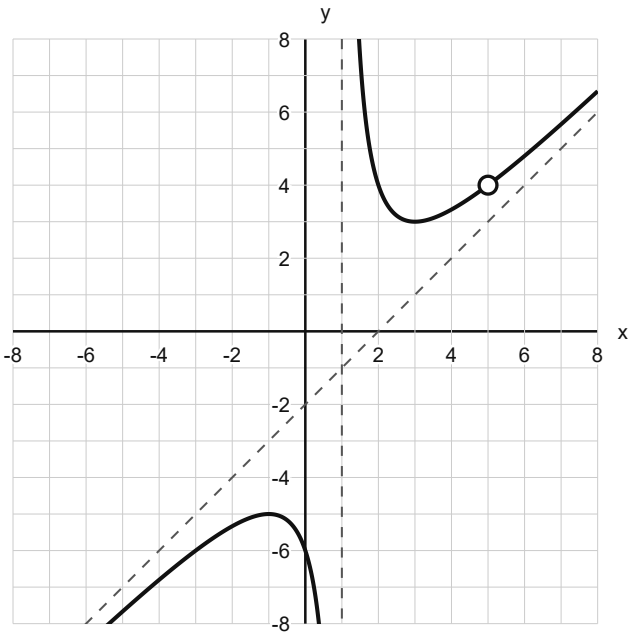
- 1: Correct factorizations, 0.25 for each original polynomial.
- 2: Valid repeated-factor cancellation 1; correct remaining fraction 1.
- 1: All original exclusions, one-third point each for -3, -2 and 3, with no spurious exclusion.

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4. 8 points

1. The parent-only generating rule is $F(x) = x - 2 + 4/(x-1)$, with $x \neq 1, 5$. Students read the graph; no calculus or generating rule is required.
2. The graph increases on $(-\infty, -1) \cup (3, 5) \cup (5, \infty)$, and decreases on $(-1, 1) \cup (1, 3)$. The missing input 5 splits the increasing branch, not its direction.
3. Relative maximum $(-1, -5)$, relative minimum $(3, 3)$. It is concave down on $(-\infty, 1)$ and concave up on $(1, 5) \cup (5, \infty)$.
4. The asymptotes are $x = 1$ and $y = x - 2$, with no horizontal line. A consistently repeated omission of the hole from increasing and concavity intervals is deducted once, not twice.



Practice rubric (8 points)

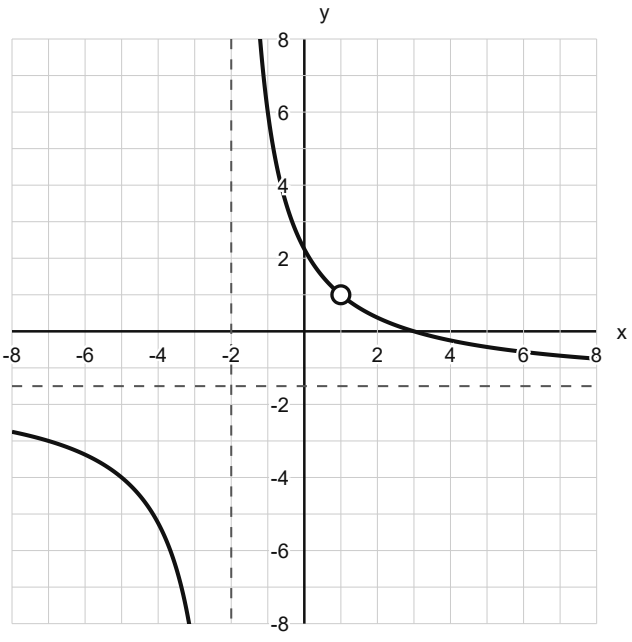
- 1: Increasing: 0.5 correct direction/turning boundaries; 0.5 correct domain partition including the hole at 5.
- 1: Decreasing: 0.5 for each of (-1,1) and (1,3); split at the vertical asymptote.
- 2: Correct maximum and minimum with type and coordinates, 1 each.
- 2: Concavity: 1 per direction with correct boundary/coverage; retain 0.5 each for correct direction if a boundary is wrong. If the sole error is the same missed hole already deducted in increasing, award that concavity partition credit.
- 2: VA x=1: 0.75; slant y=x-2: 0.75; no HA: 0.5. Credit correctly located lines separately from equation-variable mistakes.

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5. 10 points

1. Factor $R(x) = \dfrac{-3(x-1)(x-3)}{2(x+2)(x-1)} = \dfrac{-3(x-3)}{2(x+2)}$, retaining $x \neq -2, 1$.
2. The intercepts are $(3, 0)$ and $(0, 9/4)$. At the excluded input 1, the reduced rule gives $-3(1-3)/(2(1+2)) = 1$, so the hole is $(1, 1)$.
3. The vertical line is $x = -2$; the horizontal line is $y = -3/2$. Write $R(x) = -3/2 + (15/2)/(x+2)$. There is no slant asymptote.
4. The left branch is below -3/2 and falls to negative infinity near -2. The right branch decreases from positive infinity, passes through the actual intercepts, and approaches -3/2 from above. Draw an open circle at (1,1).



Practice rubric (10 points)

- 2: Factoring/cancellation 1.5; original exclusions -2 and 1: 0.5.
- 2: x-intercept (3,0) and y-intercept (0,9/4), 1 each.
- 2: Hole input 1 and exact height 1, 1 each; shown substitution supports the height.
- 2: VA x=-2 and HA y=-3/2, 1 each. For each, retain 0.5 if the correct location is demonstrated but its equation uses the wrong variable.
- 2: Sketch: consistent branches 1 (0.5 each), correctly located open hole 0.5, labeled features 0.5. Preserve follow-through and do not repeat a copied feature/equation error.

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6. 4 points

1. Division gives $S(x) = x/2 + 3/2 - 4/(2x - 2)$. Check: $(x/2 + 3/2)(2x - 2) = x^2 + 2x - 3$, leaving remainder -4.
2. Thus the slant is $y = x/2 + 3/2$, and no horizontal asymptote exists. As $x \to -\infty$, $S(x) \to -\infty$; as $x \to +\infty$, $S(x) \to +\infty$.
3. The difference $S(x) - (x/2 + 3/2) = -4/(2x - 2)$ tends to zero at both ends. It is not zero at any finite allowed input.

Practice rubric (4 points)

- 1.5: Correct quotient 1 and remainder -4/exact identity 0.5.
- 1: Correct slant line 0.75 and no-horizontal conclusion 0.25.
- 0.5: Correct far-left output direction; accept an unambiguous correctly signed comparison with the slant at negative infinity.
- 0.5: Correct far-right output direction; accept an unambiguous correctly signed comparison with the slant at positive infinity.
- 0.5: Explains that the divided remainder/difference tends to zero, not merely that the remainder was calculated.

7. 4 points

1. Rewrite $C(t) = 3 - 5/(5 + 2t)$. For every finite $t \geq 0$, the subtracted fraction is strictly positive.
2. The concentration approaches 3 grams per liter from below as more solution enters. Since $5/(5 + 2t) \neq 0$ at finite allowed times, it never equals 3 then.

Practice rubric (4 points)

- 1: Correct limit 3.
- 1: Correct grams-per-liter interpretation in context.
- 1: Correct approach from below, supported by signed algebra or equivalent valid reasoning.
- 1: Correct nonattainment conclusion with valid finite-time justification; half if conclusion alone.

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8. 4 points

1. One valid rule is $G(x) = \dfrac{-2x(x+2)}{x(x-3)}$, with $x \neq 0, 3$. Its reduced rule is $-2(x+2)/(x-3)$.
2. The remaining numerator vanishes only at -2, the remaining denominator only at 3, the leading-coefficient ratio is -2, and the canceled factor creates the only hole at 0. These statements, or equivalent labeled algebra, verify the features.
3. For this rule the missing height is $-2(0+2)/(0-3) = 4/3$. There is no y-intercept because the original function is undefined at 0, even though its reduced rule has a finite value there. Accept any rule meeting all conditions and its own correct hole height.

Practice rubric (4 points)

- 2: Valid factored construction: only required zero, pole, horizontal height and hole, 0.5 each.
- 1: Verification: original exclusions 0 and 3 with cancellation/domain distinction 0.5; explicit check of remaining zero/pole and signed leading ratio 0.5. Labeled algebra counts, prose is not required.
- 0.5: Correct hole coordinates for the chosen valid rule, with shown substitution; retain 0.25 for correct input if height is wrong.
- 0.5: No y-intercept with the original-domain explanation; 0.25 for conclusion alone.

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9. 2 points

1. Both ends approach the horizontal line $y = -2$, which is finite and nonzero. Therefore numerator and denominator degrees are equal.
2. A smaller numerator degree would give end height 0; a larger numerator degree would not give this finite horizontal end height. Exact individual degrees are not determined from the picture.

Practice rubric (2 points)

- 1: Choice C.
- 1: Connects the finite nonzero horizontal height to equal degrees; 0.5 for identifying the correct horizontal height without the degree connection.

10. 2 points

1. The double zero at 1 makes the curve touch the x-axis rather than cross. D has vertical asymptotes at $x = -2, 2$, horizontal asymptote $y = 1$, and negative values between the vertical asymptotes except at the zero.
2. A crosses at 1 and has horizontal height 0. B touches but has opposite branch signs and horizontal height -1. C has no real vertical asymptotes and has horizontal height 1/2. A correct touch description plus the horizontal height 1 distinguishes D from all three; equivalent decisive feature combinations are accepted.

Practice rubric (2 points)

- 1: Choice D.
- 0.5: Correct touching/double-zero behavior at x=1, not merely naming an intercept.
- 0.5: Another decisive supporting feature: HA height 1 or negative middle-branch sign. VAs alone do not rule out reflected B; retain 0.25 for a correct but nondecisive feature.

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11. 2 points

1. B reduces to $(x-2)(x+3)/(2(x-1))$, but retains the excluded input -1. Substitution there gives $(-3)(2)/(2(-2)) = 3/2$, so the hole is $(-1, 3/2)$.
2. The vertical asymptote is $x = 1$; division gives $y = x/2 + 1$ as the slant. These agree with the graph, as do zeros -3 and 2.
3. A fills the missing point. In C, an x+1 factor remains in the denominator after cancellation, giving a pole at -1, not a removable hole there. D reflects the curve and gives the wrong hole height sign. A common factor by itself is not enough to decide the entire graph.

Practice rubric (2 points)

- 1: Choice B.
- 0.5: Hole coordinates (-1,3/2); 0.25 for the correct input/cancellation mechanism alone. Accept a reasonable graph estimate of the height supported by the correct selected rule.
- 0.5: Another distinguishing feature, such as VA x=1, with reasoning that does not treat every common factor as automatically a hole. Together the cited features must be consistent with the whole graph.

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12. 2 points

1. The reference f has positive even leading term, so both ends go up. For $m = 8$, division by the leading x^2 gives leading behavior $4x^6$, also both ends up.
2. For $m = 7$, the leading behavior is $4x^5$, giving opposite ends. For $m = 6$, combine the tied numerator terms to get $-3x^4$ after division; for $m = 2$, the leading behavior is $-7x^4$. Both of those have ends down. Matching directions does not require matching the exact degree.

Practice rubric (2 points)

- 1: Choice A.
- 1: Correct leading sign and parity after accounting for the denominator; accept concise equivalent reasoning, not a required rejection of every other option.

13. 4 points

1. Near 1, the outputs remain finite and approach about 2.11 from both sides, supporting the hole at $x = 1$ under the stated assumptions.
2. Near -2, positive outputs grow rapidly in magnitude from both sides. Thus the likely vertical line is $x = -2$, with $H(x) \to +\infty$ as $x \to -2^-$ and as $x \to -2^+$. A vertical asymptote need not have opposite signs on its two sides.
3. One parent-only generating rule is $H(x) = 2 + 1/(x + 2)^2$, additionally excluding 1. Its missing height is $19/9$; students are not asked to find the rule or prove an exact limit from finite rounded samples.

Practice rubric (4 points)

- 1: Hole at 1.
- 1: Finite-height table evidence near 1; do not require exact 19/9.
- 1: Vertical asymptote at -2.
- 1: Growing-magnitude evidence 0.5; correct positive-infinity behavior from both sides 0.5 (0.25 each).